

SOLUTION FOR QUIZ 1

Let $A, B \subset (0, \infty)$ be nonempty and bounded sets. Set

$$\alpha = \sup A \text{ and } \beta = \sup B,$$

and define the set

$$C = \{xy : x \in A, y \in B\}.$$

Prove that

$$\sup C = \alpha\beta.$$

Proof. For any $x \in A$ we have $x \leq \alpha$, as α is an upper bound of A . Likewise, for any $y \in B$ we have $y \leq \beta$. Since $x, y > 0$, it follows that $xy \leq \alpha\beta$. Since any $z \in C$ is of the form $z = xy$, this implies that $z \leq \alpha\beta$ for all $z \in C$. Therefore, $\alpha\beta$ is an upper bound of C , which implies that

$$\sup C \leq \alpha\beta.$$

We now prove the reverse inequality. Since $xy \leq \sup C$ and $x, y > 0$, it follows that $x \leq \frac{\sup C}{y}$ for all $x \in A$. Therefore, $\frac{\sup C}{y}$ is an upper bound for A . As α is the least upper bound for A , this implies $\alpha \leq \frac{\sup C}{y}$. Rewrite this as $y \leq \frac{\sup C}{\alpha}$. This holds for every $y \in B$, so we conclude again that $\beta \leq \frac{\sup C}{\alpha}$. Hence

$$\alpha\beta \leq \sup C.$$

These two inequalities show that $\sup C = \alpha\beta$. □